



B.TECH V SEMESTER		SC	L	T	P	C
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23CS5S13	FULL STACK DEVELOPMENT - 2 (SKILL ENHANCEMENT COURSE)					

Course Objectives:

1. Make use of router, template engine and authentication using sessions to develop application in ExpressJS.
2. Build a single page application using RESTful APIs in ExpressJS.
3. Apply router and hooks in designing ReactJS application.
4. Make use of MongoDB queries to perform CRUD operations on document database.

Course Outcomes:

- CO1: Develop backend web APIs using ExpressJS by applying routing, HTTP methods, middleware, and session management..
- CO2: Develop dynamic and interactive single-page applications using ReactJS, proficiently utilizing components, state, props, event handling, routing, and modern React hooks.
- CO3: Demonstrate the use of MongoDB for managing NoSQL databases, including installation, configuration, and executing CRUD operations for data persistence.
- CO4: Analyze the integration of frontend and backend technologies in full-stack web applications, examining how web services interact with database operations

COURSE ARTICULATION MATRIX (Correlation between Cos & POs, PSOs):

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	3	2	3	-	-	-	-	-	-	2	3	3
CO2	2	2	3	2	3	-	-	-	-	-	-	2	3	3
CO3	2	2	3	2	3	-	-	-	-	-	-	-	3	3
CO4	2	2	3	2	3	-	-	-	-	-	-	-	3	3
CO5	2	2	3	2	3	-	-	-	-	-	-	2	3	3

Note: Enter Correlation Levels **1** or **2** or **3**. If there is no correlation, put **'-'** **1**- Slight (Low), **2** - Moderate (Medium), **3** - Substantial (High).



LIST OF EXPERIMENTS

1. ExpressJS – Routing, HTTP Methods, Middleware.

- a) Write a program to define a route, Handling Routes, Route Parameters, Query Parameters and URL building.
- b) Write a program to accept data, retrieve data and delete a specified resource using http methods.
- c) Write a program to show the working of middleware.

2. ExpressJS – Templating, Form Data

- a) Write a program using templating engine.
- b) Write a program to work with form data.

3. ExpressJS – Cookies, Sessions, Authentication

- a) Write a program for session management using cookies and sessions.
- b) Write a program for user authentication.

4. ExpressJS – Database, RESTful APIs

- a) Write a program to connect MongoDB database using Mongoose and perform CRUD operations.
- b) Write a program to develop a single page application using RESTful APIs.

5. ReactJS – Render HTML, JSX, Components – function & Class

- a) Write a program to render HTML to a web page.
- b) Write a program for writing markup with JSX.
- c) Write a program for creating and nesting components (function and class).

6. ReactJS – Props and States, Styles, Respond to Events

- a) Write a program to work with props and states.
- b) Write a program to add styles (CSS & Sass Styling) and display data.
- c) Write a program for responding to events.

7. ReactJS – Conditional Rendering, Rendering Lists, React Forms

- a) Write a program for conditional rendering.
- b) Write a program for rendering lists.
- c) Write a program for working with different form fields using react forms.

8. ReactJS – React Router, Updating the Screen

- a) Write a program for routing to different pages using react router.
- b) Write a program for updating the screen.

9. ReactJS – Hooks, Sharing data between Components

- a) Write a program to understand the importance of using hooks.



- b) Write a program for sharing data between components.

10. MongoDB – Installation, Configuration, CRUD operations

- a) Install MongoDB and configure ATLAS
- b) Write MongoDB queries to perform CRUD operations on document using insert(), find(), update(), remove()

11. MongoDB – Databases, Collections and Records

- a) Write MongoDB queries to Create and drop databases and collections.
- b) Write MongoDB queries to work with records using find(), limit(), sort(), createIndex(), aggregate().

12. Augmented Programs: (Any 2 must be completed)

- a) Design a to-do list application using NodeJS and ExpressJS.
- b) Design a Quiz app using ReactJS.

Complete the MongoDB certification from MongoDB University website.

References:

1. Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node, Vasan Subramanian, 2nd edition, APress, O'Reilly.
2. Node.js in Action, Mike Cantelon, Mark Harter, T.J. Holowaychuk, Nathan Rajlich, Manning Publications. (Chapters 1-11)
3. React Quickly, AzatMardan, Manning Publications(Chapters 1-8,12-14)

Web Links:

1. ExpressJS - <https://www.tutorialspoint.com/expressjs>
2. ReactJS - <https://www.w3schools.com/REACT> (and) <https://react.dev/learn#>
3. MongoDB - <https://learn.mongodb.com/learning-paths/introduction-to-mongodb>